

# DAY – 2 HW (Loops Practice)

## Questions

1. Print numbers from **1 to 10** using a loop.
2. Print numbers from **10 to 1** in reverse order.
3. Print all **even numbers from 1 to 50**.
4. Print all **odd numbers from 1 to 50**.
5. Print the **multiplication table** of a number entered by the user.
6. Find the **sum of first N natural numbers**.
7. Find the **factorial of a number**.
8. Count digits in a number.
9. Reverse a number.

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## Solutions

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### 1) Print Numbers from 1 to 10

#### Explanation

A loop starts from 1 and increases by 1 each time until it reaches 10.

#### C++

```
for(int i=1;i<=10;i++)  
    cout<<i<<" ";
```

#### Java

```
for(int i=1;i<=10;i++)  
    System.out.print(i+" ");
```

#### Python

```
for i in range(1,11):  
    print(i, end=" ")
```

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### 2) Print Numbers from 10 to 1

#### Explanation

Start from 10 and decrease the value each iteration until it becomes 1.

## C++

```
for(int i=10;i>=1;i--)  
    cout<<i<<" ";
```

## Java

```
for(int i=10;i>=1;i--)  
    System.out.print(i+" ");
```

## Python

```
for i in range(10,0,-1):  
    print(i, end=" ")
```

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## 3) Print Even Numbers from 1 to 50

### Explanation

Even numbers are divisible by 2. Start from 2 and increase by 2 each time.

## C++

```
for(int i=2;i<=50;i+=2)  
    cout<<i<<" ";
```

## Java

```
for(int i=2;i<=50;i+=2)  
    System.out.print(i+" ");
```

## Python

```
for i in range(2,51,2):  
    print(i, end=" ")
```

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## 4) Print Odd Numbers from 1 to 50

### Explanation

Odd numbers are not divisible by 2. Start from 1 and increase by 2.

## C++

```
for(int i=1;i<=50;i+=2)  
    cout<<i<<" ";
```

## Java

```
for(int i=1;i<=50;i+=2)
    System.out.print(i+" ");
```

## Python

```
for i in range(1,51,2):
    print(i, end=" ")
```

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## 5) Multiplication Table

### Explanation

User gives a number. Loop runs from 1 to 10 and multiplies the number each time.

Example: For input 5 → print 5×1, 5×2, etc.

### C++

```
int n;
cin>>n;

for(int i=1;i<=10;i++)
    cout<<n<<" x "<<i<<" = "<<n*i<<endl;
```

### Java

```
int n = sc.nextInt();

for(int i=1;i<=10;i++)
    System.out.println(n+" x "+i+" = "+(n*i));
```

### Python

```
n=int(input())

for i in range(1,11):
    print(f"{n} x {i} = {n*i}")
```

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## 6) Sum of First N Natural Numbers

### Explanation

We keep adding numbers from 1 to N in a variable called `sum`.

Example: N=5 → 1+2+3+4+5 = 15.

### C++

```
int n,sum=0;
```

```
cin>>n;

for(int i=1;i<=n;i++)
    sum+=i;

cout<<sum;
```

## Java

```
int n=sc.nextInt(), sum=0;

for(int i=1;i<=n;i++)
    sum+=i;

System.out.println(sum);
```

## Python

```
n=int(input())
sum=0

for i in range(1,n+1):
    sum+=i

print(sum)
```

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# 7) Factorial of a Number

## Explanation

Factorial means multiplying all numbers from 1 to N.

Example:

$$5! = 1 \times 2 \times 3 \times 4 \times 5$$

## C++

```
int n,fact=1;
cin>>n;

for(int i=1;i<=n;i++)
    fact*=i;

cout<<fact;
```

## Java

```
int n=sc.nextInt(), fact=1;

for(int i=1;i<=n;i++)
    fact*=i;

System.out.println(fact);
```

## Python

```
n=int(input())
fact=1

for i in range(1,n+1):
    fact*=i

print(fact)
```

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## 8) Count Digits in Number

### Explanation

Each time we remove the last digit by dividing number by 10 and increase count.

Example:

1234 → 123 → 12 → 1 → 0 (4 steps)

### C++

```
int n,count=0;
cin>>n;

while(n>0){
    count++;
    n/=10;
}

cout<<count;
```

### Java

```
int n=sc.nextInt(), count=0;

while(n>0){
    count++;
    n/=10;
}

System.out.println(count);
```

## Python

```
n=int(input())
count=0

while n>0:
    count+=1
    n//=10

print(count)
```

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## 9) Reverse a Number

### Explanation

Take last digit and add it to reversed number.

Example:

1234  $\rightarrow$  4  $\rightarrow$  43  $\rightarrow$  432  $\rightarrow$  4321

### C++

```
int n, rev=0;
cin>>n;

while(n>0){
    rev=rev*10+n%10;
    n/=10;
}

cout<<rev;
```

### Java

```
int n=sc.nextInt(), rev=0;

while(n>0){
    rev=rev*10+n%10;
    n/=10;
}

System.out.println(rev);
```

### Python

```
n=int(input())
rev=0

while n>0:
    rev=rev*10+n%10
    n//=10

print(rev)
```